

Numerical Algorithms: Problem Set 1 — Complete Solutions

With Detailed Explanations for Exam Preparation

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1 Numerical Differentiation (Q1–Q10)

Key idea: We approximate $f'(x)$ using nearby function evaluations. Two competing errors always exist: *truncation error* (decreases as $h \rightarrow 0$) and *rounding/cancellation error* (increases as $h \rightarrow 0$).

Q1 (Easy, Short Answer) — Forward Difference Definition

Define the forward-difference approximation to $f'(x)$ using step size $h > 0$. State the leading-order truncation error in big- O notation as $h \rightarrow 0$, assuming f is sufficiently smooth.

Solution: The forward-difference approximation is:

$$D_f(h) = \frac{f(x+h) - f(x)}{h}.$$

Derivation of truncation error: Taylor-expand $f(x+h)$ about x :

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + O(h^3).$$

Subtracting $f(x)$ and dividing by h :

$$D_f(h) = f'(x) + \frac{h}{2}f''(x) + O(h^2).$$

The leading truncation error term is $\frac{h}{2}f''(x)$, so the truncation error is $\mathcal{O}(h)$ as $h \rightarrow 0$.

Q2 (Moderate, Long Answer) — Forward Difference Computation

Let $f(x) = x^2 + e^x + \ln x + \sin x$ (domain $x > 0$).

(a) Compute $f'(x)$ analytically.

Solution: Differentiating term by term:

$$f'(x) = 2x + e^x + \frac{1}{x} + \cos x.$$

At $x = 0.5$:

$$f'(0.5) = 2(0.5) + e^{0.5} + \frac{1}{0.5} + \cos(0.5) = 1 + 1.648721 + 2 + 0.877583 = \mathbf{5.526304}.$$

(b) At $x = 0.5$, compute $D_f(h) = \frac{f(x+h)-f(x)}{h}$ for $h = 10^{-2}$ and $h = 10^{-5}$.

Solution: First compute $f(0.5)$:

$$\begin{aligned} f(0.5) &= 0.25 + e^{0.5} + \ln(0.5) + \sin(0.5) \\ &= 0.25 + 1.648721 - 0.693147 + 0.479426 = 1.685000. \end{aligned}$$

For $h = 10^{-2}$: $f(0.51) = (0.51)^2 + e^{0.51} + \ln(0.51) + \sin(0.51)$

$$= 0.2601 + 1.665299 - 0.673345 + 0.488177 = 1.740231.$$

$$D_f(10^{-2}) = \frac{1.740231 - 1.685000}{0.01} = \frac{0.055231}{0.01} = \mathbf{5.5231}.$$

For $h = 10^{-5}$: $f(0.50001) \approx 1.685000 + 10^{-5} \cdot f'(0.5) \approx 1.685000 + 5.5263 \times 10^{-5} = 1.685055.$

$$D_f(10^{-5}) = \frac{1.685055 - 1.685000}{10^{-5}} = \frac{0.000055263}{10^{-5}} \approx \mathbf{5.5263}.$$

h	$D_f(h)$	Error vs. exact
10^{-2}	5.5231	7.3×10^{-3}
10^{-5}	5.5263	$\sim 10^{-5}$
Exact	5.5263038	0

Halving h by a factor of 1000 roughly reduces the error by 1000, consistent with $\mathcal{O}(h)$ convergence.

(c) Briefly explain why decreasing h does not necessarily improve the estimate indefinitely.

Solution: When h is very small, $f(x+h) \approx f(x)$, and their difference suffers **catastrophic cancellation** in floating-point arithmetic: the numerator $f(x+h) - f(x)$ loses significant digits. The rounding error in $D_f(h)$ scales as $\sim \varepsilon_m/h$, which *increases* as $h \rightarrow 0$. There is an optimal $h^* \sim \sqrt{\varepsilon_m} \approx 10^{-8}$ that balances truncation and rounding errors.

Q3 (Difficult, Long Answer) — Optimal Step Size

Forward-difference error model: $E(h) \approx c_1 h + c_2 \varepsilon_m / h$.

(a) Find the optimal h in terms of ε_m .

Solution: Minimize $E(h) = c_1 h + c_2 \varepsilon_m h^{-1}$ over $h > 0$:

$$\frac{dE}{dh} = c_1 - \frac{c_2 \varepsilon_m}{h^2} = 0 \implies h^2 = \frac{c_2 \varepsilon_m}{c_1} \implies h^* = \sqrt{\frac{c_2 \varepsilon_m}{c_1}}.$$

For $c_1 \sim c_2 \sim 1$: $h^* \sim \sqrt{\varepsilon_m}$.

(b) Find the corresponding optimal achievable accuracy $E(h^*)$.

Solution: Substitute $h^* = \sqrt{c_2 \varepsilon_m / c_1}$ back:

$$E(h^*) = c_1 \sqrt{\frac{c_2 \varepsilon_m}{c_1}} + c_2 \varepsilon_m \cdot \sqrt{\frac{c_1}{c_2 \varepsilon_m}} = \sqrt{c_1 c_2 \varepsilon_m} + \sqrt{c_1 c_2 \varepsilon_m} = 2\sqrt{c_1 c_2 \varepsilon_m}.$$

So $E(h^*) \sim \sqrt{\varepsilon_m}$.

Interpretation: No matter how carefully we choose h , the forward-difference method can only achieve accuracy $\sim \sqrt{\varepsilon_m}$. To do better requires a higher-order method (e.g., centered difference, which achieves $\sim \varepsilon_m^{2/3}$).

Q4 (Moderate, Short Answer) — Numerical Example of Optimal h

Using $\varepsilon_m = 10^{-16}$, $c_1 \sim c_2 \sim 1$: estimate h^* and $E(h^*)$.

Solution:

$$h^* \sim \sqrt{\varepsilon_m} = \sqrt{10^{-16}} = 10^{-8}.$$

$$E(h^*) \sim \sqrt{\varepsilon_m} = 10^{-8}.$$

Practical implication: In double-precision arithmetic, you should use $h \approx 10^{-8}$ for forward differences, and you can expect about 8 correct decimal digits in the derivative estimate.

Q5 (Moderate, Long Answer) — Central Difference Derivation

Derive the central-difference approximation $D_c(h) = \frac{f(x+h/2) - f(x-h/2)}{h}$ and show the leading truncation term is $\mathcal{O}(h^2)$.

Solution: Taylor-expand about x :

$$f\left(x + \frac{h}{2}\right) = f(x) + \frac{h}{2}f'(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f''(x) + \frac{1}{6}\left(\frac{h}{2}\right)^3 f'''(x) + \cdots$$

$$f\left(x - \frac{h}{2}\right) = f(x) - \frac{h}{2}f'(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f''(x) - \frac{1}{6}\left(\frac{h}{2}\right)^3 f'''(x) + \cdots$$

Subtracting the second from the first:

$$f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right) = hf'(x) + \frac{h^3}{24}f'''(x) + \mathcal{O}(h^5).$$

Dividing by h :

$$D_c(h) = f'(x) + \frac{h^2}{24}f'''(x) + \mathcal{O}(h^4).$$

The leading truncation error is $\frac{h^2}{24}f'''(x) = \mathcal{O}(h^2)$ —**one order better than forward difference**.

Key insight: The even-order terms ($f'', f^{(4)}, \dots$) cancel out by symmetry, leaving only odd powers of h .

Q6 (Moderate, Long Answer) — Central Difference on a Polynomial

Let $p(x) = 5x^3 + 4x^2 + 3x + 2$.

(a) Compute $p'(x)$ analytically.

Solution: $p'(x) = 15x^2 + 8x + 3$.

(b) Expand $p(x \pm h/2)$ and compute $D_c(h)$.

Solution: Compute each term of $p(x + h/2) - p(x - h/2)$:

$$(x + \frac{h}{2})^3 - (x - \frac{h}{2})^3 = 3x^2h + \frac{h^3}{4},$$

$$(x + \frac{h}{2})^2 - (x - \frac{h}{2})^2 = 2xh,$$

$$(x + \frac{h}{2}) - (x - \frac{h}{2}) = h.$$

Therefore:

$$p(x + \frac{h}{2}) - p(x - \frac{h}{2}) = 5(3x^2h + \frac{h^3}{4}) + 4(2xh) + 3h = 15x^2h + \frac{5h^3}{4} + 8xh + 3h.$$

$$D_c(h) = 15x^2 + \frac{5h^2}{4} + 8x + 3 = p'(x) + \frac{5h^2}{4}.$$

Note: $D_c(h) = p'(x) + \frac{5h^2}{4}$, which equals $p'(x)$ only when $h = 0$. The formula is exact for polynomials of degree ≤ 2 (where $p''' = 0$). For this cubic, the h^2 -error term comes from the cubic part: $\frac{h^2}{24}p'''(x) = \frac{h^2}{24} \cdot 30 = \frac{5h^2}{4}$. Consistent with Q5.

(c) Explain why floating-point arithmetic may fail for extremely small h .

Solution: For very small h , $p(x+h/2)$ and $p(x-h/2)$ become almost equal in floating-point. Their difference $p(x+h/2) - p(x-h/2)$ suffers catastrophic cancellation, losing many significant bits. Even though $D_c(h)$ is algebraically exact for $p' + \mathcal{O}(h^2)$, the numerical subtraction $p(x+h/2) - p(x-h/2)$ is computed with only a few correct bits when h is tiny. The relative rounding error in the final result scales as $\sim \varepsilon_m/h$, eventually dominating.

Q7 (Difficult, Long Answer) — Higher-Order Difference Formula

The formula:

$$D_4(h) = \frac{8f(x + h/4) + f(x - h/2) - 8f(x - h/4) - f(x + h/2)}{3h}.$$

(a) Show the $\mathcal{O}(h^2)$ term cancels and the leading truncation error is $\mathcal{O}(h^4)$.

Solution: Recognize D_4 as Richardson extrapolation on D_c . Let:

$$\begin{aligned} D_c(h) &= \frac{f(x + h/2) - f(x - h/2)}{h} = f'(x) + \frac{h^2}{24}f'''(x) + O(h^4), \\ D_c(h/2) &= \frac{f(x + h/4) - f(x - h/4)}{h/2} = f'(x) + \frac{(h/2)^2}{24}f'''(x) + O(h^4) = f'(x) + \frac{h^2}{96}f'''(x) + O(h^4). \end{aligned}$$

Note that the numerator of D_4 equals:

$$8[f(x + h/4) - f(x - h/4)] - [f(x + h/2) - f(x - h/2)] = 8 \cdot D_c(h/2) \cdot (h/2) - D_c(h) \cdot h.$$

$$\text{So } D_4(h) = \frac{4D_c(h/2) - D_c(h)}{3} \text{ (Richardson extrapolation to cancel the } h^2 \text{ term!).}$$

Cancellation:

$$D_4(h) = \frac{4 \left[f'(x) + \frac{h^2}{96}f''' + O(h^4) \right] - \left[f'(x) + \frac{h^2}{24}f''' + O(h^4) \right]}{3} = \frac{3f'(x) + \frac{h^2}{24}f''' - \frac{h^2}{24}f''' + O(h^4)}{3} = f'(x) + \mathcal{O}(h^4).$$

The $\mathcal{O}(h^2)$ terms cancel exactly. D_4 achieves $\mathcal{O}(h^4)$ accuracy.

(b) Discuss when $D_4(h)$ is superior to $D_c(h)$ in practice, and when it may not be.

Solution: **When D_4 is superior:** When the step h is not too small and truncation error dominates. The error of D_4 scales as h^4 while D_c scales as h^2 —a dramatic improvement for moderate h values.

When D_4 may not be superior:

1. **Very small h :** D_4 uses 4 function evaluations with smaller spacings ($h/4$ and $h/2$). More subtractions of nearly-equal quantities amplifies rounding error faster. The optimal h for D_4 is smaller but the floor of rounding-limited accuracy is similar.
2. **Noisy functions:** If f has stochastic noise larger than ε_m , higher-order stencils amplify noise more aggressively.
3. **Expensive evaluations:** D_4 uses 4 evaluations vs. 2 for D_c . If f is cheap, no issue; if expensive, it doubles the cost.

Q8 (Moderate, Long Answer) — Second Derivative Formula

$$f''(x) \approx \frac{f(x + h) - 2f(x) + f(x - h)}{h^2}.$$

(a) Show the truncation error order.

Solution: Add the Taylor expansions:

$$\begin{aligned}f(x+h) &= f + hf' + \frac{h^2}{2}f'' + \frac{h^3}{6}f''' + \frac{h^4}{24}f^{(4)} + \dots \\f(x-h) &= f - hf' + \frac{h^2}{2}f'' - \frac{h^3}{6}f''' + \frac{h^4}{24}f^{(4)} + \dots\end{aligned}$$

Adding: $f(x+h) + f(x-h) = 2f + h^2f'' + \frac{h^4}{12}f^{(4)} + \dots$

Rearranging:

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{h^2}{12}f^{(4)}(x) + O(h^4).$$

Truncation error: $\mathcal{O}(h^2)$.

(b) Propose an error model with round-off, and discuss the optimal h .

Solution: Rounding error: Each of $f(x+h)$, $f(x)$, $f(x-h)$ has absolute floating-point error $\sim \varepsilon_m |f|$. After the weighted sum $(1, -2, 1)$ and division by h^2 , the rounding error is approximately $\frac{4\varepsilon_m |f|}{h^2}$.

Total error model:

$$E(h) \approx \frac{h^2}{12}|f^{(4)}| + \frac{4\varepsilon_m |f|}{h^2}.$$

Optimal h : Minimize $\phi(h) = Ah^2 + B\varepsilon_m/h^2$:

$$\frac{d\phi}{dh} = 2Ah - \frac{2B\varepsilon_m}{h^3} = 0 \implies h^4 = \frac{B\varepsilon_m}{A} \implies h_{\text{opt}} \sim \varepsilon_m^{1/4}.$$

For $\varepsilon_m = 10^{-16}$: $h_{\text{opt}} \sim 10^{-4}$. This is larger than for first derivatives because the h^2 in the denominator amplifies rounding more steeply.

Q9 (Easy, Short Answer) — Second Derivative from First Differences

Give a concise explanation of “compute the second derivative by calculating the first derivative twice.” Show the key algebraic step.

Solution: Conceptual explanation: Apply the finite-difference operator Δ_h (which approximates d/dx) twice. First, $\Delta_h[f](x) = \frac{f(x+h)-f(x)}{h}$ approximates $f'(x)$. Then apply Δ_h again to f' :

$$\Delta_h[f'](x) \approx \frac{f'(x+h) - f'(x)}{h} \approx \frac{1}{h} \left[\frac{f(x+2h) - f(x+h)}{h} - \frac{f(x+h) - f(x)}{h} \right] = \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}.$$

Key algebraic step: A difference of first differences becomes a second-order difference:

$$\Delta_h[\Delta_h[f]](x) = \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}.$$

The centered version uses backward and forward differences symmetrically:

$$f''(x) \approx \frac{1}{h} \left[\frac{f(x+h) - f(x)}{h} - \frac{f(x) - f(x-h)}{h} \right] = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}.$$

Q10 (Difficult, Long Answer) — Adaptive Step Size

Design an adaptive rule that updates h using $h_{\text{new}} = h \left(\frac{\varepsilon_R}{2\varepsilon_T} \right)^{1/3}$.

Solution: Background: For centered difference $D_c(h)$, the error is:

$$E(h) \approx \underbrace{\varepsilon_T}_{\sim Ch^2} + \underbrace{\varepsilon_R/h}_{\text{rounding}}.$$

The optimal h satisfies $\varepsilon_R/h = 2Ch^2 = 2\varepsilon_T$, i.e., $h = (\varepsilon_R/(2C))^{1/3} = h(\varepsilon_R/(2\varepsilon_T))^{1/3}$ when $\varepsilon_T = Ch^2$.

Algorithm:

(i) Estimate ε_T (truncation error): Compute $D_c(h)$ and $D_c(h/2)$. Since $D_c(h) \approx f'(x) + Ch^2$ and $D_c(h/2) \approx f'(x) + Ch^2/4$:

$$D_c(h) - D_c(h/2) \approx \frac{3}{4}Ch^2 = \frac{3}{4}\varepsilon_T(h).$$

So estimate $\varepsilon_T \approx \frac{4}{3}|D_c(h) - D_c(h/2)|$.

(ii) Estimate ε_R (rounding error): The rounding error in $D_c(h)$ is approximately $\varepsilon_R \approx 2\varepsilon_m|f(x)|$, where $\varepsilon_m \approx 10^{-16}$ is machine epsilon. Estimate using $\varepsilon_R \approx 2\varepsilon_m \max(|f(x+h/2)|, |f(x-h/2)|)$.

(iii) Update h and terminate:

$$h_{\text{new}} = h \left(\frac{\varepsilon_R}{2\varepsilon_T} \right)^{1/3}.$$

Repeat until $h_{\text{new}} \approx h$ (convergence) or until $\varepsilon_T \approx \varepsilon_R$ (balance achieved). Terminate when $|\varepsilon_T - \varepsilon_R| < \delta$ for some tolerance δ , or after a fixed number of iterations.

Failure modes:

- Near zeros of f :** If $f(x) \approx 0$, our estimate $\varepsilon_R \approx 2\varepsilon_m|f|$ is near zero, so the algorithm drives h too small, amplifying truncation error.
- Near inflection points ($f''' \approx 0$):** The truncation error estimate $\varepsilon_T \approx 0$, causing $h_{\text{new}} \rightarrow \infty$ (blow-up).
- Highly oscillatory or discontinuous f :** The difference $D_c(h) - D_c(h/2)$ may not reflect the true truncation error structure.

2 Numerical Integration (Q11–Q22)

Q11 (Moderate, Short Answer) — Composite Trapezoidal Rule

Write the composite trapezoidal rule for $\int_a^b f(x) dx$ using N subintervals.

Solution: With step size $h = (b - a)/N$ and nodes $x_i = a + ih$ for $i = 0, \dots, N$:

$$T_N = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{N-1} f(x_i) + f(x_N) \right] = h \left[\frac{1}{2}f(a) + f(x_1) + f(x_2) + \dots + f(x_{N-1}) + \frac{1}{2}f(b) \right].$$

Global truncation error: For sufficiently smooth f ,

$$\left| \int_a^b f \, dx - T_N \right| \leq \frac{(b-a)h^2}{12} \max_{x \in [a,b]} |f''(x)| = \mathcal{O}(h^2).$$

Q12 (Moderate, Long Answer) — Trapezoidal Rule Computation

Compute a trapezoidal approximation of $\int_0^{1.2} \left(x - x^2 + x^3 - x^4 + \frac{\sin(13x)}{13} \right) dx$ using $h = 0.2$.

(a) List the grid points.

Solution: $h = 0.2$, so $N = 6$. Grid: $x_0 = 0$, $x_1 = 0.2$, $x_2 = 0.4$, $x_3 = 0.6$, $x_4 = 0.8$, $x_5 = 1.0$, $x_6 = 1.2$.

(b) Write the weighted sum and (c) compute the numerical value.

Solution: Let $g(x) = x - x^2 + x^3 - x^4 + \sin(13x)/13$. Values (using a calculator):

x_i	$\sin(13x_i)$ (rad)	$g(x_i)$
0.0	$\sin(0) = 0$	0.000000
0.2	$\sin(2.6) \approx 0.51550$	$0.1664 + 0.03965 = 0.20605$
0.4	$\sin(5.2) \approx -0.87758$	$0.2784 - 0.06751 = 0.21089$
0.6	$\sin(7.8) \approx 0.99878$	$0.3264 + 0.07683 = 0.40323$
0.8	$\sin(10.4) \approx -0.82736$	$0.2624 - 0.06364 = 0.19876$
1.0	$\sin(13) \approx 0.41982$	$0 + 0.03230 = 0.03230$
1.2	$\sin(15.6) \approx 0.10785$	$-0.5856 + 0.00830 = -0.57730$

Trapezoidal rule with $h = 0.2$:

$$\begin{aligned} T_6 &= \frac{0.2}{2} [g_0 + 2g_1 + 2g_2 + 2g_3 + 2g_4 + 2g_5 + g_6] \\ &= 0.1 [0 + 2(0.20605) + 2(0.21089) + 2(0.40323) + 2(0.19876) + 2(0.03230) + (-0.57730)] \\ &= 0.1 [0 + 0.41210 + 0.42178 + 0.80646 + 0.39752 + 0.06460 - 0.57730] \\ &= 0.1 \times 1.52516 = \mathbf{0.15252}. \end{aligned}$$

Exact value:

$$\int_0^{1.2} (x - x^2 + x^3 - x^4) \, dx = \left[\frac{x^2}{2} - \frac{x^3}{3} + \frac{x^4}{4} - \frac{x^5}{5} \right]_0^{1.2} = 0.72 - 0.576 + 0.5184 - 0.49766 = 0.16474.$$

$$\int_0^{1.2} \frac{\sin(13x)}{13} \, dx = \left[-\frac{\cos(13x)}{169} \right]_0^{1.2} = \frac{-\cos(15.6) + 1}{169} = \frac{0.9942 + 1}{169} \approx 0.01180.$$

Exact $\approx 0.16474 + 0.01180 = \mathbf{0.17654}$.

Error $= |0.17654 - 0.15252| \approx 0.024$. The large error is due to the rapidly oscillatory $\sin(13x)/13$ term: with frequency 13 rad/unit and $h = 0.2$, there are only ~ 1 point per period, making the trapezoidal rule very inaccurate for this component. A much smaller h (say $h = 0.01$) would significantly improve accuracy.

Q13 (Moderate, Long Answer) — Derive Simpson's Rule

Derive Simpson's rule on $[x, x + 2h]$ by fitting a quadratic through the three equally spaced points.

Solution: Let $x_0 = x$, $x_1 = x + h$, $x_2 = x + 2h$. Fit the quadratic Lagrange interpolant through (x_0, f_0) , (x_1, f_1) , (x_2, f_2) . Substituting $t = x_0 + sh$ (so $s \in [0, 2]$):

$$p(s) = f_0 \frac{(s-1)(s-2)}{(-1)(-2)} + f_1 \frac{s(s-2)}{(1)(-1)} + f_2 \frac{s(s-1)}{(2)(1)}.$$

Integrate $\int_0^2 p(s) h ds$:

$$\begin{aligned} \int_0^2 \frac{(s-1)(s-2)}{2} ds &= \frac{1}{2} \int_0^2 (s^2 - 3s + 2) ds = \frac{1}{2} \left[\frac{s^3}{3} - \frac{3s^2}{2} + 2s \right]_0^2 = \frac{1}{2} \left(\frac{8}{3} - 6 + 4 \right) = \frac{1}{3}, \\ \int_0^2 \frac{s(s-2)}{-1} ds &= - \int_0^2 (s^2 - 2s) ds = - \left[\frac{s^3}{3} - 2s \right]_0^2 = \frac{4}{3}, \\ \int_0^2 \frac{s(s-1)}{2} ds &= \frac{1}{2} \int_0^2 (s^2 - s) ds = \frac{1}{2} \left[\frac{s^3}{3} - \frac{s^2}{2} \right]_0^2 = \frac{1}{3}. \end{aligned}$$

Therefore:

$$\int_x^{x+2h} p(t) dt = h \left(\frac{f_0}{3} + \frac{4f_1}{3} + \frac{f_2}{3} \right) = \frac{h}{3} [f(x) + 4f(x+h) + f(x+2h)].$$

Simpson's rule: $\int_x^{x+2h} f(t) dt \approx \frac{h}{3} [f_0 + 4f_1 + f_2]$.

Q14 (Difficult, Long Answer) — Simpson's Rule Error Analysis

(a) Show the leading neglected term involves $f^{(4)}$.

Solution: On the single panel $[x, x + 2h]$, the quadratic interpolant p_2 agrees with f at 3 points. The error $f - p_2$ is:

$$f(t) - p_2(t) = \frac{f^{(3)}(\xi)}{3!} (t - x_0)(t - x_1)(t - x_2) \quad (\text{3rd-order Lagrange remainder}).$$

However, when we *integrate* this over $[x, x + 2h]$, the integral of $(t - x_0)(t - x_1)(t - x_2)$ is zero by odd symmetry about $x_1 = x + h$ (the function is antisymmetric). So the $O(h^3)$ term vanishes, and the next term from the 4th-order remainder gives:

$$\int_x^{x+2h} [f(t) - p_2(t)] dt = \frac{f^{(4)}(\xi)}{4!} \int_x^{x+2h} (t - x_0)(t - x_1)^2(t - x_2) dt + \dots$$

This integral is $-h^5/15$, so the leading error term involves $f^{(4)}$.

(b) The single-panel error is $\mathcal{O}(h^5)$.

Solution: The exact error for a single Simpson panel is:

$$E_{\text{single}} = -\frac{h^5}{90} f^{(4)}(\xi) \quad \text{for some } \xi \in (x, x + 2h).$$

So the single-panel error is $\mathcal{O}(h^5)$. Compared to the trapezoid's $\mathcal{O}(h^3)$ single-panel error, Simpson's is much more accurate per panel.

(c) The composite Simpson's rule has global error $\mathcal{O}(h^4)$.

Solution: For $L = b - a$ fixed and n panels (with n even), each of width $2h$ (so $n/2$ applications of the rule):

$$|E_{\text{total}}| = \frac{n}{2} \cdot \frac{h^5}{90} |f^{(4)}(\xi)| \approx \frac{L}{2h} \cdot \frac{h^5}{90} |f^{(4)}| = \frac{Lh^4}{180} |f^{(4)}| = \mathcal{O}(h^4).$$

Two orders better than the composite trapezoidal rule!

Q15 (Easy, Short Answer) — Evaluation Count Comparison

Composite Simpson with $h = 10^{-3}$ over $[0, 1.2]$: how many function evaluations? Compare to Gauss with $n = 21$ points.

Solution: Composite Simpson on $[0, 1.2]$ with $h = 10^{-3}$: Number of subintervals: $N = 1.2/10^{-3} = 1200$ (must be even). Number of nodes: $N + 1 = \mathbf{1201}$ function evaluations.

Gauss-Legendre with $n = 21$ points: exactly **21** function evaluations, achieving polynomial exactness up to degree $2(21) - 1 = 41$.

Comparison: Gauss uses $\approx 57\times$ fewer evaluations. For smooth integrands, Gauss is far more efficient. However, if the integrand is piecewise smooth, oscillatory, or has singularities, the adaptive composite approach may be preferable.

Q16 (Moderate, Long Answer) — Two-Point Gauss-Legendre Derivation

(a) Set up the system from exactness for $1, x, x^2, x^3$.

Solution: Let nodes t_1, t_2 and weights w_1, w_2 satisfy $\int_{-1}^1 p(t) dt = w_1 p(t_1) + w_2 p(t_2)$ for $p = 1, t, t^2, t^3$:

$$p = 1 : \quad w_1 + w_2 = 2 \tag{1}$$

$$p = t : \quad w_1 t_1 + w_2 t_2 = 0 \tag{2}$$

$$p = t^2 : \quad w_1 t_1^2 + w_2 t_2^2 = \frac{2}{3} \tag{3}$$

$$p = t^3 : \quad w_1 t_1^3 + w_2 t_2^3 = 0 \tag{4}$$

(b) Use symmetry to solve.

Solution: By symmetry of $[-1, 1]$: set $t_1 = -t$, $t_2 = t$ with $t > 0$, and $w_1 = w_2 = w$.

From (1): $2w = 2 \Rightarrow w = 1$. From (2): $w(-t) + w(t) = 0$. From (3): $w t^2 + w t^2 = 2t^2 = \frac{2}{3} \Rightarrow t^2 = \frac{1}{3} \Rightarrow t = \frac{1}{\sqrt{3}}$. From (4): $w(-t)^3 + w(t)^3 = 0$.

Result: Nodes $t_{1,2} = \pm \frac{1}{\sqrt{3}}$, weights $w_1 = w_2 = 1$.

(c) Conclude the rule integrates any cubic exactly on $[-1, 1]$.

Solution: Any cubic $p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ is a linear combination of $\{1, t, t^2, t^3\}$. Since the rule is exact for each basis polynomial (equations (1)–(4)), by linearity it is exact for all cubics. The degree of exactness is **3**.

Q17 (Difficult, Long Answer) — Three-Point Gauss–Legendre

Derive the three-point rule by imposing exactness for $1, x, x^2, x^3, x^4, x^5$.

(a) Use symmetry.

Solution: By symmetry of $[-1, 1]$, set $x_2 = 0$, $x_1 = -x_3$ with $x_3 > 0$, and $w_1 = w_3$. We have 3 unknowns: $x_3, w_1 = w_3, w_2$.

(b) Solve for the unknowns.

Solution: Apply exactness conditions:

$$\begin{aligned} p = 1 : \quad 2w_1 + w_2 &= 2 \quad \Rightarrow \quad w_2 = 2 - 2w_1. \\ p = t^2 : \quad 2w_1x_3^2 + w_2 \cdot 0 &= \frac{2}{3} \quad \Rightarrow \quad w_1 = \frac{1}{3x_3^2}. \\ p = t^4 : \quad 2w_1x_3^4 &= \frac{2}{5} \quad \Rightarrow \quad w_1x_3^4 = \frac{1}{5}. \end{aligned}$$

From the last two: $w_1x_3^4 = \frac{1}{5}$ and $w_1 = \frac{1}{3x_3^2}$, so $\frac{x_3^4}{3x_3^2} = \frac{x_3^2}{3} = \frac{1}{5}$:

$$x_3^2 = \frac{3}{5} \Rightarrow x_3 = \sqrt{\frac{3}{5}}.$$

Then $w_1 = \frac{1}{3 \cdot (3/5)} = \frac{1}{9/5} = \frac{5}{9}$, and $w_2 = 2 - 2 \cdot \frac{5}{9} = 2 - \frac{10}{9} = \frac{8}{9}$.

Three-point Gauss–Legendre rule:

Node t_i	Weight w_i
$-\sqrt{3/5} \approx -0.77460$	$5/9 \approx 0.5556$
0	$8/9 \approx 0.8889$
$+\sqrt{3/5} \approx +0.77460$	$5/9 \approx 0.5556$

(c) Verify exactness for three monomials.

Solution: $p = 1$: $(5/9) + (8/9) + (5/9) = 18/9 = 2 = \int_{-1}^1 1 \, dt$

$p = t^2$: $2 \cdot (5/9) \cdot (3/5) + 0 = 2 \cdot (1/3) = 2/3 = \int_{-1}^1 t^2 \, dt$

$p = t^4$: $2 \cdot (5/9) \cdot (9/25) + 0 = 2 \cdot (1/9) \cdot (9/5) = 2/5 = \int_{-1}^1 t^4 \, dt$

Odd monomials vanish by symmetry. The rule is exact for degree $\leq 2(3) - 1 = 5$.

Q18 (Moderate, Long Answer) — Gauss–Legendre on $[a, b]$

(a) Derive the affine mapping $x = \frac{b-a}{2}t + \frac{a+b}{2}$.

Solution: We need $x : [-1, 1] \rightarrow [a, b]$, i.e., $x(-1) = a$ and $x(1) = b$. The unique affine map is:

$$x(t) = \frac{b-a}{2}t + \frac{a+b}{2}, \quad dx = \frac{b-a}{2} dt.$$

Substituting into the integral:

$$\int_a^b f(x) dx = \frac{b-a}{2} \int_{-1}^1 f\left(\frac{b-a}{2}t + \frac{a+b}{2}\right) dt.$$

(b) Use the 2-point G-L rule to approximate $I = \int_0^2 e^x dx$.

Solution: Here $a = 0$, $b = 2$: $x(t) = t + 1$, $dx = dt$.

Mapped nodes: $x_1 = -1/\sqrt{3} + 1 \approx 0.42265$ and $x_2 = 1/\sqrt{3} + 1 \approx 1.57735$.

$$I \approx \frac{2-0}{2} [e^{x_1} + e^{x_2}] = 1 \cdot [e^{0.42265} + e^{1.57735}] \approx 1.52607 + 4.84237 = \mathbf{6.36844}.$$

(c) Compare to exact value $e^2 - 1$.

Solution: $e^2 - 1 \approx 7.38906 - 1 = 6.38906$.

Absolute error = $|6.38906 - 6.36844| \approx \mathbf{0.0206}$.

This is very accurate for just 2 function evaluations. The 2-point rule is exact for cubics; since e^x is smooth and well-approximated by low-degree polynomials on $[0, 2]$, the result is excellent.

Q19 (Easy, Short Answer) — Gauss Degree of Exactness

An n -point Gauss-Legendre rule is exact for polynomials up to degree $2n - 1$. For $n = 5$?

Solution: For $n = 5$: degree of exactness = $2(5) - 1 = \mathbf{9}$.

The 5-point rule integrates all polynomials of degree ≤ 9 exactly.

Degree-10 polynomials: Not guaranteed. The degree of exactness is 9; a degree-10 polynomial will generally give an error. The 5-point rule cannot represent degree-10 polynomials exactly because it has only 10 free parameters (5 nodes + 5 weights), which are fully determined by the degree ≤ 9 exactness conditions.

Q20 (Difficult, Long Answer) — Integrating Near Singularities

Consider $I = \int_0^1 x^{-1/2} dx$.

(a) Explain why uniform fixed-step rules behave poorly near $x = 0$.

Solution: $f(x) = x^{-1/2}$ has a singularity at $x = 0$: $f(0) = +\infty$. Uniform rules like the trapezoidal rule assume f is smooth and bounded. Near $x = 0$:

- The function changes extremely rapidly: $f'(x) = -\frac{1}{2}x^{-3/2} \rightarrow -\infty$.
- Linear (or polynomial) interpolation is a poor approximation for $x^{-1/2}$ near 0.
- The error bound for trapezoidal rule involves $\max |f''|$, which is $+\infty$ here.

Even though $I = [2\sqrt{x}]_0^1 = 2$ is finite, the trapezoidal estimate converges *very slowly* with small h near the singularity.

(b) Propose two remedies.

Solution:

1. **Change of variables:** Transform the integral to remove the singularity. Use a substitution that “stretches” the singularity and makes the integrand smooth.
2. **Adaptivity/non-uniform meshing:** Use much smaller subintervals near $x = 0$ (where the function varies rapidly) and larger ones away from 0. An adaptive quadrature algorithm will automatically refine near the singularity.

(c) Apply the substitution $x = u^2$.

Solution: Let $x = u^2$, so $dx = 2u du$. When $x = 0$: $u = 0$; when $x = 1$: $u = 1$.

$$I = \int_0^1 x^{-1/2} dx = \int_0^1 (u^2)^{-1/2} \cdot 2u du = \int_0^1 u^{-1} \cdot 2u du = \int_0^1 2 du = 2.$$

The singularity is completely eliminated. The transformed integrand is the constant $g(u) = 2$, which any quadrature rule integrates exactly!

General principle: If $f(x) \sim Cx^{-\alpha}$ near $x = 0$ with $0 < \alpha < 1$, the substitution $x = u^{1/(1-\alpha)}$ maps the singularity to a smooth (in fact, C^1) integrand.

Q21 (Easy, Short Answer) — Homemade vs. Professional Code

List two pros and two cons of “homemade code” versus “public/professional code” for numerical algorithms.

Solution: Homemade code:

- + **Full control/customizability:** Can tailor precision, data structures, and stopping criteria to the specific problem.
- + **Educational value:** Writing code from scratch deepens understanding of the algorithm.
- **Time-consuming:** Implementing, debugging, and validating from scratch is expensive.
- **Less robust:** Professional libraries have been tested on thousands of edge cases; homemade code may fail on corner cases.

Professional/public code:

- + Battle-tested, fast, and highly optimized.
- + Well-documented with known convergence properties and error bounds.
- Black-box behavior: harder to diagnose subtle numerical issues.
- May be overkill or inflexible for highly specialized problems.

Q22 (Moderate, Short Answer) — Central-Difference Utility Behavior

A numerical derivative utility uses a central-difference formula. Error behaves like $\mathcal{O}(h^2) + \mathcal{O}(\varepsilon_m/h)$.

(a) Explain why the empirically best h is often 10^{-5} to 10^{-6} in double precision.

Solution: For centered differences, truncation error $\sim Ch^2$ and rounding error $\sim \varepsilon_m/h$. The optimal h balances these:

$$2Ch = \varepsilon_m/h^2 \implies h^3 = \frac{\varepsilon_m}{2C} \implies h_{\text{opt}} \sim \varepsilon_m^{1/3}.$$

With $\varepsilon_m \approx 10^{-16}$ and $C \sim 1$: $h_{\text{opt}} \sim (10^{-16})^{1/3} = 10^{-16/3} \approx 10^{-5.33}$. This explains why $h \approx 10^{-5}$ to 10^{-6} is empirically optimal for double precision centered differences.

(b) Explain why higher-order finite-difference stencils do not always lead to dramatic improvements in practice.

Solution: Higher-order stencils (like D_4 in Q7) have smaller truncation errors, but:

1. **More evaluations:** A 4-point stencil uses more cancellation of nearly-equal values, amplifying rounding error.
2. **Lower optimal h :** The optimal h for an $O(h^4)$ method is smaller than for $O(h^2)$, but the rounding-error floor improves only modestly.
3. **Function noise:** If f has even tiny amounts of noise (measurement, model error), the higher-order method amplifies it more.
4. **Practical bottleneck:** Often the bottleneck is not truncation error but rounding, and all methods hit the same rounding-error floor $\sim \varepsilon_m^{2/3}$ (for centered) or $\sim \varepsilon_m^{3/4}$ (for 4th-order), which differ by only a fraction of a decimal place.

3 Convexity and Optimization Foundations (Q23–Q30)

Q23 — Convex Sets

(a) State the definition of a convex set $C \subseteq \mathbb{R}^d$ in terms of line segments.

Solution: A set $C \subseteq \mathbb{R}^d$ is **convex** if for every $x, y \in C$ and every $\theta \in [0, 1]$:

$$\theta x + (1 - \theta)y \in C.$$

Equivalently, the line segment joining any two points in C lies entirely within C .

(b) Prove that the intersection of two convex sets is convex.

Solution: Claim: If A and B are convex, then $A \cap B$ is convex.

Proof: Let $x, y \in A \cap B$ and $\theta \in [0, 1]$.

- Since $x, y \in A$ (both in A) and A is convex: $\theta x + (1 - \theta)y \in A$.
- Since $x, y \in B$ (both in B) and B is convex: $\theta x + (1 - \theta)y \in B$.

Therefore $\theta x + (1 - \theta)y \in A \cap B$. $\square$

Generalization: The intersection of *any* family of convex sets is convex (by the same argument applied to each set in the family).

Q24 — Convex Hull Computation

Let $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 2)$ in $\mathbb{R}^2$.

(a) Write the convex hull $\text{co}(x_1, x_2, x_3)$ as a set of convex combinations.

Solution: $\text{co}(x_1, x_2, x_3) = \{\theta_1(0, 0) + \theta_2(1, 0) + \theta_3(0, 2) : \theta_i \geq 0, \theta_1 + \theta_2 + \theta_3 = 1\} = \{(\theta_2, 2\theta_3) : \theta_2 \geq 0, \theta_3 \geq 0, \theta_2 + \theta_3 \leq 1\} = \{(s, t) : s \geq 0, t \geq 0, s + t/2 \leq 1\}$, which is the filled triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 2)$.

(b) Determine whether $z = (0.25, 0.5)$ lies in the convex hull.

Solution: We need $\theta_1, \theta_2, \theta_3 \geq 0$ with $\theta_1 + \theta_2 + \theta_3 = 1$ such that: $\theta_2 = 0.25$ and $2\theta_3 = 0.5 \Rightarrow \theta_3 = 0.25$. Then $\theta_1 = 1 - 0.25 - 0.25 = 0.5 \geq 0$.

Yes, $z = (0.25, 0.5)$ lies in the convex hull, with coefficients $(\theta_1, \theta_2, \theta_3) = (0.5, 0.25, 0.25)$.

(c) Geometric interpretation.

Solution: $z = (0.25, 0.5)$ lies inside the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 2)$. Geometrically, z is a weighted average of the three vertices (with weights $1/2$, $1/4$, $1/4$), which intuitively lies closer to $x_1 = (0, 0)$ (largest weight) and equidistant between x_2 and x_3 .

Q25 — Halfspace as a Cone

The halfspace $H = \{x \in \mathbb{R}^d : a^\top x \leq b\}$ is convex. Under what condition on b does H become a cone?

Solution: A cone must satisfy: for all $x \in H$ and all $\alpha > 0$: $\alpha x \in H$.

$$\alpha x \in H \iff a^\top(\alpha x) \leq b \iff \alpha(a^\top x) \leq b.$$

For this to hold for *all* $\alpha > 0$ and *all* x with $a^\top x \leq b$:

- If $a^\top x < 0$: $\alpha(a^\top x) \leq b$ for all $\alpha > 0$ as long as $b \geq 0$.
- If $a^\top x = 0$: $\alpha \cdot 0 = 0 \leq b$ requires $b \geq 0$.
- If $0 < a^\top x \leq b$ (possible when $b > 0$): for large α , $\alpha(a^\top x) > b$. Fails!

Conclusion: H is a cone if and only if $\boxed{b = 0}$. In that case $H = \{x : a^\top x \leq 0\}$, and scaling by any $\alpha > 0$ keeps $a^\top(\alpha x) = \alpha a^\top x \leq 0$.

Q26 — Convex Functions and Jensen's Inequality

(a) State the definition of a convex function $f : \mathbb{R}^d \rightarrow \mathbb{R}$.

Solution: f is **convex** if $\text{dom}(f)$ is a convex set and for all $x, y \in \text{dom}(f)$ and $\theta \in [0, 1]$:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

The function value at any convex combination is at most the convex combination of function values – the graph of f lies *below* the chord.

(b) Define midpoint convexity.

Solution: f is **midpoint convex** if for all $x, y \in \text{dom}(f)$:

$$f\left(\frac{x + y}{2}\right) \leq \frac{f(x) + f(y)}{2}.$$

This is the special case $\theta = 1/2$ of the convexity definition. Midpoint convexity is weaker than full convexity.

(c) Prove (or outline a proof) that continuous + midpoint convex $\Rightarrow$ convex.

Solution: **Step 1:** By repeated application of midpoint convexity, establish convexity at all *dyadic rationals* $\theta = k/2^n$ ($k = 0, \dots, 2^n$):

For $n = 1$: midpoint convexity gives $f(\frac{x+y}{2}) \leq \frac{f(x)+f(y)}{2}$ (i.e., $\theta = 1/2$).

For general n : by induction, $f(\frac{kx+(2^n-k)y}{2^n}) \leq \frac{kf(x)+(2^n-k)f(y)}{2^n}$ for all integers k .

Step 2: Every $\theta \in [0, 1]$ is the limit of dyadic rationals $\theta_n \rightarrow \theta$.

Step 3: By continuity of f , the inequality $f(\theta_n x + (1 - \theta_n)y) \leq \theta_n f(x) + (1 - \theta_n)f(y)$ passes to the limit as $\theta_n \rightarrow \theta$:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

This proves full convexity. $\square$

Q27 — First-Order Characterization of Convexity

(a) State the first-order condition: f is convex iff $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$ for all x, y .

Solution: This says the *tangent hyperplane* at any point is a global lower bound for f . Geometrically: the graph of f lies above all its tangent planes. This is both necessary and sufficient for convexity when f is differentiable on a convex domain.

(b) Use this condition to argue $f(x) = \|x\|_2$ is convex on $\mathbb{R}^d$.

Solution: We can argue directly from the definition: for $\theta \in [0, 1]$,

$$\|\theta x + (1 - \theta)y\|_2 \leq \|\theta x\|_2 + \|(1 - \theta)y\|_2 = \theta\|x\|_2 + (1 - \theta)\|y\|_2$$

by the triangle inequality and positive homogeneity. So f is convex.

Using the first-order condition (for $x \neq 0$): $\nabla f(x) = x/\|x\|_2$. We need $\|x\|_2 \geq \|y\|_2 + \frac{y^\top}{\|y\|_2}(x - y)$ for $y \neq 0$:

$$\|x\|_2 \geq \frac{y^\top x}{\|y\|_2} \iff \|x\|_2 \|y\|_2 \geq y^\top x,$$

which holds by Cauchy–Schwarz.

(c) Is $\sin(x)$ convex on $[0, \pi]$? Show if the first-order condition is violated.

Solution: $(\sin x)'' = -\sin x \leq 0$ on $[0, \pi]$ (since $\sin x \geq 0$). So $\sin(x)$ is **concave** (not convex) on $[0, \pi]$.

Violation example: Take $y = \pi/4$, $x = \pi/2$:

$$f(x) = \sin(\pi/2) = 1.$$

$$f(y) + f'(y)(x - y) = \sin(\pi/4) + \cos(\pi/4)(\pi/2 - \pi/4) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{\pi}{4} \approx 0.707 + 0.555 = 1.262.$$

Since $1 < 1.262$, we have $f(x) < f(y) + \nabla f(y)(x - y)$, violating the first-order convexity condition. (confirms concavity).

Q28 — Norms

(a) State the three norm axioms.

Solution: A function $\|\cdot\| : \mathbb{R}^d \rightarrow \mathbb{R}$ is a **norm** if:

1. **Definiteness/positive definiteness:** $\|x\| \geq 0$ for all x , and $\|x\| = 0 \iff x = 0$.
2. **Positive homogeneity:** $\|\alpha x\| = |\alpha|\|x\|$ for all $\alpha \in \mathbb{R}$ and $x \in \mathbb{R}^d$.
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbb{R}^d$.

(b) Prove that any norm is a convex function.

Solution: For $\theta \in [0, 1]$:

$$\|\theta x + (1 - \theta)y\| \stackrel{\text{(iii)}}{\leq} \|\theta x\| + \|(1 - \theta)y\| \stackrel{\text{(ii)}}{=} \theta\|x\| + (1 - \theta)\|y\|.$$

Used: triangle inequality (iii), then positive homogeneity (ii) with $|\theta| = \theta$ (since $\theta \geq 0$). This is exactly the definition of convexity. $\square$

(c) Verify that $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$ is a norm for $p \geq 1$.

Solution:

1. **Definiteness:** $\|x\|_p = 0 \iff \sum |x_i|^p = 0 \iff \text{each } |x_i| = 0 \iff x = 0$.
2. **Homogeneity:** $\|\alpha x\|_p = (\sum |\alpha x_i|^p)^{1/p} = |\alpha|(\sum |x_i|^p)^{1/p} = |\alpha|\|x\|_p$.
3. **Triangle inequality:** This is **Minkowski's inequality**: $\|x + y\|_p \leq \|x\|_p + \|y\|_p$ for $p \geq 1$. (standard result; proof uses Hölder's inequality).

Q29 — Dual Norms and Hölder's Inequality

Dual norm: $\|u\|_* = \sup\{u^\top x : \|x\| \leq 1\}$.

(a) Prove $\|\cdot\|_*$ is a norm.

Solution: Definiteness: $\|u\|_* \geq 0$ (sup of non-negative quantities if we include $x = 0$). If $u \neq 0$: take $x = u/\|u\|$; then $u^\top x = \|u\| > 0$, so $\|u\|_* \geq \|u\|/\|u\| \cdot \|u\| > 0$. Actually: $u^\top (u/\|u\|_2) \cdot \|u\|_2/\|u\|_2 \dots$ Let me just note: if $u \neq 0$ then choosing $x = u/\|u\|_2$ (or any x with $u^\top x > 0$) gives $\|u\|_* > 0$.

Homogeneity: $\|\alpha u\|_* = \sup\{\alpha u^\top x : \|x\| \leq 1\} = |\alpha| \sup\{u^\top x : \|x\| \leq 1\} = |\alpha|\|u\|_*$ (for $\alpha < 0$, the sup flips).

Triangle inequality:

$$\|u + v\|_* = \sup_{\|x\| \leq 1} (u + v)^\top x \leq \sup_{\|x\| \leq 1} u^\top x + \sup_{\|x\| \leq 1} v^\top x = \|u\|_* + \|v\|_*. \quad \checkmark$$

(b) Prove the generalized Hölder inequality $u^\top x \leq \|u\|_* \|x\|$.

Solution: If $\|x\| = 0$: both sides are 0.

If $\|x\| > 0$: let $\hat{x} = x/\|x\|$, so $\|\hat{x}\| = 1$. Then:

$$u^\top \hat{x} \leq \sup_{\|y\| \leq 1} u^\top y = \|u\|_*.$$

Multiplying both sides by $\|x\|$:

$$u^\top x = u^\top (\hat{x}\|x\|) = (u^\top \hat{x})\|x\| \leq \|u\|_* \|x\|. \quad \square$$

(c) Compute the dual of $\|\cdot\|_1$ and the dual of $\|\cdot\|_\infty$.

Solution: Dual of $\|\cdot\|_1$:

$$\|u\|_{1,*} = \sup\{u^\top x : \|x\|_1 \leq 1\} = \sup\{\sum_i u_i x_i : \sum_i |x_i| \leq 1\}.$$

The maximum is achieved by putting all weight on the i with largest $|u_i|$: $\|u\|_{1,*} = \max_i |u_i| = \|u\|_\infty$.

Dual of $\|\cdot\|_\infty$:

$$\|u\|_{\infty,*} = \sup\{u^\top x : \|x\|_\infty \leq 1\} = \sup\{\sum_i u_i x_i : |x_i| \leq 1 \forall i\}.$$

Maximize by choosing $x_i = \text{sign}(u_i)$: $\|u\|_{\infty,*} = \sum_i |u_i| = \|u\|_1$.

Summary: $(\|\cdot\|_1)^* = \|\cdot\|_\infty$ and $(\|\cdot\|_\infty)^* = \|\cdot\|_1$. More generally, $(\|\cdot\|_p)^* = \|\cdot\|_q$ where $1/p + 1/q = 1$.

Q30 — Schur Complement via Partial Minimization

Let $C \succ 0$ and $Z = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} \succeq 0$. Define $L(x, y) = \begin{pmatrix} x \\ y \end{pmatrix}^\top Z \begin{pmatrix} x \\ y \end{pmatrix}$.

(a) Expand $L(x, y)$.

Solution:

$$L(x, y) = x^\top Ax + 2x^\top By + y^\top Cy.$$

(The factor 2 comes from the two off-diagonal blocks B and $B^\top$.)

(b) Minimize $L(x, y)$ over y for fixed x .

Solution: Set $\nabla_y L = 0$:

$$\nabla_y L = 2B^\top x + 2Cy = 0 \implies y^*(x) = -C^{-1}B^\top x.$$

(Valid since $C \succ 0$, hence invertible.)

(c) Substitute $y^*(x)$ back to show the minimized value equals $x^\top (A - BC^{-1}B^\top)x$.

Solution:

$$\begin{aligned} L(x, y^*(x)) &= x^\top Ax + 2x^\top B(-C^{-1}B^\top x) + (-C^{-1}B^\top x)^\top C(-C^{-1}B^\top x) \\ &= x^\top Ax - 2x^\top BC^{-1}B^\top x + x^\top BC^{-1} \underbrace{C \cdot C^{-1}}_{=I} B^\top x \\ &= x^\top Ax - 2x^\top BC^{-1}B^\top x + x^\top BC^{-1}B^\top x \\ &= x^\top (A - BC^{-1}B^\top)x. \end{aligned}$$

The expression $A - BC^{-1}B^\top$ is the **Schur complement** of C in Z .

(d) Conclude that $A - BC^{-1}B^\top \succeq 0$.

Solution: Since $Z \succeq 0$, the quadratic form $L(x, y) \geq 0$ for all (x, y) . In particular, for any fixed x , $\inf_y L(x, y) \geq 0$.

But we showed $\inf_y L(x, y) = L(x, y^*(x)) = x^\top (A - BC^{-1}B^\top)x$.

Therefore $x^\top (A - BC^{-1}B^\top)x \geq 0$ for all $x \in \mathbb{R}^n$, which means:

$$A - BC^{-1}B^\top \succeq 0. \quad \square$$

Summary: $Z \succeq 0$ with $C \succ 0$ implies $A - BC^{-1}B^\top \succeq 0$ (the Schur complement is positive semidefinite).

Q(i) — Gradient Descent Update Rule

Write the gradient descent update rule and state what information is needed at each iteration.

Solution: The **gradient descent update** is:

$$x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)}),$$

where $\eta > 0$ is the step size (learning rate).

Information needed at each iteration: The gradient $\nabla f(x^{(k)})$ evaluated at the current iterate. (For second-order methods like Newton's, you would also need the Hessian $\nabla^2 f(x^{(k)})$.)

Q(j) — Three Explicit GD Steps on a Quadratic

$$f(x) = \frac{1}{2}x^\top Qx - b^\top x, \quad Q = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \eta = 0.1.$$

(a) Show $\nabla f(x) = Qx - b$.

Solution: For symmetric Q :

$$\nabla f(x) = \frac{1}{2}(Q + Q^\top)x - b = Qx - b,$$

since $Q = Q^\top$ (verify: $Q_{12} = Q_{21} = 1$).

(b) Starting from $x^{(0)} = \mathbf{0}$, compute $x^{(1)}, x^{(2)}, x^{(3)}$.

Solution: **Step 1:** $\nabla f(x^{(0)}) = Q \cdot 0 - b = -b = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$.

$$x^{(1)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.1 \begin{pmatrix} -1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix}.$$

$$\text{Step 2: } Qx^{(1)} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.7 \end{pmatrix}. \quad \nabla f(x^{(1)}) = \begin{pmatrix} 0.6 \\ 0.7 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -0.4 \\ -1.3 \end{pmatrix}.$$

$$x^{(2)} = \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix} - 0.1 \begin{pmatrix} -0.4 \\ -1.3 \end{pmatrix} = \begin{pmatrix} 0.14 \\ 0.33 \end{pmatrix}.$$

$$\text{Step 3: } Qx^{(2)} = \begin{pmatrix} 4(0.14) + 0.33 \\ 0.14 + 3(0.33) \end{pmatrix} = \begin{pmatrix} 0.89 \\ 1.13 \end{pmatrix}. \quad \nabla f(x^{(2)}) = \begin{pmatrix} 0.89 \\ 1.13 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -0.11 \\ -0.87 \end{pmatrix}.$$

$$x^{(3)} = \begin{pmatrix} 0.14 \\ 0.33 \end{pmatrix} - 0.1 \begin{pmatrix} -0.11 \\ -0.87 \end{pmatrix} = \begin{pmatrix} 0.151 \\ 0.417 \end{pmatrix}.$$

(c) Compute $f(x^{(k)})$ for $k = 0, 1, 2, 3$ and comment on monotonic decrease.

Solution: $f(x) = \frac{1}{2}x^\top Qx - b^\top x$.

$$f(x^{(0)}) = 0 - 0 = \mathbf{0}.$$

$f(x^{(1)})$: $\frac{1}{2}[0.1, 0.2]Q[0.1, 0.2]^\top = \frac{1}{2}[0.1, 0.2] \cdot [0.6, 0.7]^\top = \frac{1}{2}(0.20) = 0.10$. $b^\top x^{(1)} = 0.1 + 0.4 = 0.5$.
So $f(x^{(1)}) = 0.10 - 0.50 = \mathbf{-0.400}$.

$f(x^{(2)})$: $\frac{1}{2}[0.14, 0.33] \cdot [0.89, 1.13]^\top = \frac{1}{2}(0.4975) = 0.249$. $b^\top x^{(2)} = 0.14 + 0.66 = 0.80$. $f(x^{(2)}) = 0.249 - 0.80 = \mathbf{-0.551}$.

$f(x^{(3)})$: $\frac{1}{2}[0.151, 0.417] \cdot [1.021, 1.402]^\top = \frac{1}{2}(0.739) = 0.369$. $b^\top x^{(3)} = 0.151 + 0.834 = 0.985$.
 $f(x^{(3)}) = 0.369 - 0.985 = \mathbf{-0.616}$.

k	$x^{(k)}$	$f(x^{(k)})$
0	(0, 0)	0
1	(0.100, 0.200)	-0.400
2	(0.140, 0.330)	-0.551
3	(0.151, 0.417)	-0.616

Monotonic decrease: Yes, f decreases at each step: $0 > -0.400 > -0.551 > -0.616$.

The optimal solution is $x^* = Q^{-1}b = \frac{1}{11} \begin{pmatrix} 3 & -1 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} 1 \\ 7 \end{pmatrix} \approx (0.091, 0.636)$ with $f(x^*) = -\frac{15}{22} \approx -0.682$. The iterates are converging, but slowly (the step size $\eta = 0.1$ is conservative for this problem).

Q(k) — Three Steps with Backtracking Line Search (Armijo)

$$f(x) = \frac{1}{2}\|Ax - y\|_2^2 \text{ with } A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}, y = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, x^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Backtracking: start $\eta = 1$, shrink by $\beta = 1/2$; Armijo condition $f(x - \eta \nabla f) \leq f(x) - c\eta \|\nabla f\|^2$, $c = 10^{-4}$.

(a) Derive $\nabla f(x) = A^\top(Ax - y)$.

Solution: $f(x) = \frac{1}{2}(Ax - y)^\top(Ax - y)$. $\nabla f(x) = A^\top(Ax - y)$. (Standard result: $\nabla_x \frac{1}{2}\|Ax - y\|^2 = A^\top(Ax - y)$.)

$$\text{Here: } A^\top = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

(b) Perform three iterations showing accepted stepsizes.

Solution: Iteration 1: $x^{(0)} = (0, 0)^\top$. $Ax^{(0)} - y = -y = (-1, 0)^\top$. $\nabla f = A^\top(-1, 0)^\top = (2, 1)^\top \cdot (-1) + (0, 1)^\top \cdot 0 \dots$

$$\text{Actually: } A^\top = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}. \nabla f(x^{(0)}) = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}.$$

$$f(x^{(0)}) = \frac{1}{2}\|(-1, 0)\|^2 = \frac{1}{2}. \|\nabla f\|^2 = 4.$$

Test $\eta = 1$: $x = (0, 0) - 1 \cdot (-2, 0) = (2, 0)$. $Ax - y = (4, 2) - (1, 0) = (3, 2)$. $f = \frac{1}{2}(9 + 4) = 6.5$. Armijo: $6.5 \leq 0.5 - 10^{-4}(4) = 0.4996$? **No**.

Test $\eta = 0.5$: $x = (1, 0)$. $Ax - y = (2, 1) - (1, 0) = (1, 1)$. $f = 1$. Armijo: $1 \leq 0.5 - 0.0002 = 0.4998$? **No**.

Test $\eta = 0.25$: $x = (0.5, 0)$. $Ax - y = (1, 0.5) - (1, 0) = (0, 0.5)$. $f = \frac{1}{2}(0.25) = 0.125$. Armijo: $0.125 \leq 0.5 - 0.0001 = 0.4999$? **Yes**. Accept $\eta = 0.25$.

$$x^{(1)} = (0.5, 0)^\top.$$

Iteration 2: $Ax^{(1)} - y = (1, 0.5) - (1, 0) = (0, 0.5)^\top$. $\nabla f(x^{(1)}) = A^\top(0, 0.5)^\top = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$. $f(x^{(1)}) = 0.125$. $\|\nabla f\|^2 = 0.5$.

Test $\eta = 1$: $x = (0.5, 0) - (0.5, 0.5) = (0, -0.5)$. $Ax - y = (0, -0.5) - (1, 0) = (-1, -0.5)$. $f = \frac{1}{2}(1.25) = 0.625$. Armijo: $0.625 \leq 0.125 - 5 \times 10^{-5} = 0.124995$? **No**.

Test $\eta = 0.5$: $x = (0.5, 0) - 0.5(0.5, 0.5) = (0.25, -0.25)$. $Ax - y = (0.5, 0) - (1, 0) = (-0.5, 0)$. $f = 0.125$. Armijo: $0.125 \leq 0.125 - 0.000025$? **No** (barely).

Test $\eta = 0.25$: $x = (0.5, 0) - 0.25(0.5, 0.5) = (0.375, -0.125)$. $Ax - y = (0.75, 0.25) - (1, 0) = (-0.25, 0.25)$. $f = \frac{1}{2}(0.125) = 0.0625$. Armijo: $0.0625 \leq 0.125 - 0.0000125$? **Yes**. Accept $\eta = 0.25$.

$$x^{(2)} = (0.375, -0.125)^\top.$$

Iteration 3: $Ax^{(2)} - y = (0.75, 0.25) - (1, 0) = (-0.25, 0.25)$. $\nabla f(x^{(2)}) = A^\top(-0.25, 0.25)^\top = \begin{pmatrix} -0.25 \\ 0.25 \end{pmatrix}$. $f(x^{(2)}) = 0.0625$. $\|\nabla f\|^2 = 0.125$.

Test $\eta = 1$: $x = (0.375, -0.125) - (-0.25, 0.25) = (0.625, -0.375)$. $Ax - y = (1.25, 0.25) - (1, 0) = (0.25, 0.25)$. $f = 0.0625$. Armijo: $0.0625 \leq 0.0625 - 10^{-4}(0.5)(0.125) = 0.0625 - 6.25 \times 10^{-6}$? **No**.

Test $\eta = 0.5$: $x = (0.375, -0.125) + 0.5(0.25, -0.25) = (0.5, -0.25)$. $Ax - y = (1, 0.25) - (1, 0) = (0, 0.25)$. $f = \frac{1}{2}(0.0625) = 0.03125$. Armijo: $0.03125 \leq 0.0625 - 3.125 \times 10^{-6} \approx 0.0625$? **Yes**. Accept $\eta = 0.5$.

$$x^{(3)} = (0.5, -0.25)^\top.$$

Summary: Accepted stepsizes: $\eta_1 = 0.25$, $\eta_2 = 0.25$, $\eta_3 = 0.5$.

k	$x^{(k)}$	$f(x^{(k)})$	Accepted η
0	(0, 0)	0.5000	—
1	(0.5, 0)	0.1250	0.25
2	(0.375, -0.125)	0.0625	0.25
3	(0.5, -0.25)	0.0313	0.50

Monotonic decrease: $0.5 > 0.125 > 0.0625 > 0.0313$. The exact minimizer is $x^* = (A^\top A)^{-1} A^\top y$.

(c) Briefly explain why line search can help when a constant stepsize is unstable.

Solution: A fixed step size η that is too large can cause gradient descent to overshoot the minimum and diverge: if $\eta > 2/\lambda_{\max}(A^\top A)$, the iterates oscillate or grow unboundedly. Backtracking line search adaptively reduces η until a sufficient decrease condition (Armijo) is met, guaranteeing convergence regardless of the initial η . This is especially valuable when the curvature of f varies across iterations (e.g., near a narrow valley), so no single fixed η works globally.